SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	RAVURU PRANATHI	**Reg. No.:**	192525076
Date:	24-07-2026	**Duration:**	60 minutes

Code of Conduct

*I **R. Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: R. Pranathi

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
    - Explain the problem in your own words.
    - Identify the objectives of the proposed system.
    - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
    - Identify all the stakeholders involved in the system.
    - Explain the roles and responsibilities of each stakeholder.
    - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
    - Describe the existing system or current process.
    - Identify its strengths and limitations.
    - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
    - Design a software-based solution for the given problem.
    - Describe the major functional modules of the proposed system.
    - Identify the functional and non-functional requirements.
    - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
   - Justify why the selected methodology/framework is suitable for the proposed system.
   - Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5

Case Study 36 – SDG 9: Smart Construction Project Management System

1.Understand the Problem Statement

Problem Explanation

Construction companies currently manage multiple projects using spreadsheets, paper documents, phone calls, and manual reports. This leads to poor communication, delayed procurement of materials, lack of real-time progress tracking, budget overruns, and missed project deadlines. A centralized Smart Construction Project Management System is required to automate planning, resource allocation, project monitoring, and reporting.

Objectives of the Proposed System

- Centralize all construction project information.
- Track project progress in real time.
- Manage materials and equipment efficiently.
- Improve communication among stakeholders.
- Reduce project delays and cost overruns.
- Generate automatic reports and dashboards.
- Improve project quality and decision-making.

Scope of the Problem

The proposed system covers:

- Project planning
- Task scheduling
- Resource management
- Material procurement
- Budget monitoring
- Employee management
- Progress tracking
- Report generation
- Notification system
- Dashboard for project monitoring

2. Identify Key Stakeholders and Challenges

Stakeholder	Roles & Responsibilities	Challenges
Company Management	Monitor all projects and budgets	Lack of project visibility
Project Manager	Plan and supervise construction	Delays and poor coordination
Site Engineer	Update work progress	Manual reporting
Procurement Team	Purchase materials	Late procurement
Workers	Perform assigned tasks	Communication gaps
Suppliers	Deliver construction materials	Delivery delays
Clients	Monitor project status	Lack of transparency
System Administrator	Manage users and security	Data security and maintenance

Major Challenges

- Delayed material procurement
- Poor communication
- Manual reporting errors
- Budget overruns
- Lack of centralized data
- Difficulty tracking project progress
- Resource conflicts
- Poor decision making

3. Analyze the Current Approach

Existing System

Construction companies currently use:

- Excel spreadsheets
- Paper records
- Emails
- Phone calls
- WhatsApp communication
- Manual reports

Strengths

- Low initial cost
- Easy to start
- Familiar to employees

Weaknesses

- No centralized database
- Duplicate information
- Human errors
- Slow communication
- Difficult report generation
- No live tracking
- Security issues

Problems/Gaps

- Project delays
- Material shortages
- Budget mismanagement
- Poor resource allocation
- Lack of accountability
- Time-consuming reporting

4. Proposed Software Solution Using Software Engineering Principles

Proposed System

Develop a **Smart Construction Project Management System**, a centralized web/cloud application that manages all construction activities digitally.

Functional Modules

1. User Login & Authentication
2. Project Management
3. Task Scheduling
4. Employee Management
5. Material Procurement
6. Resource Allocation
7. Budget Management
8. Progress Tracking
9. Report Generation
10. Notification & Alerts

Functional Requirements

- User registration and login
- Create and manage projects
- Assign tasks
- Track project progress
- Manage materials
- Generate reports
- Dashboard visualization
- Email/SMS notifications

Non-Functional Requirements

- High performance
- Scalability
- Security
- Reliability
- Availability
- Easy-to-use interface
- Maintainability
- Backup and recovery

Software Engineering Principles

Modularity

- Independent modules for projects, resources, procurement, and reporting.

Scalability

- Supports multiple projects and users simultaneously.

Maintainability

- Easy software updates and bug fixes.

Reliability

- Accurate data processing with automatic backups.

Usability

- Simple and user-friendly interface.

Security

- Role-based access control, encrypted passwords, secure database, audit logs.

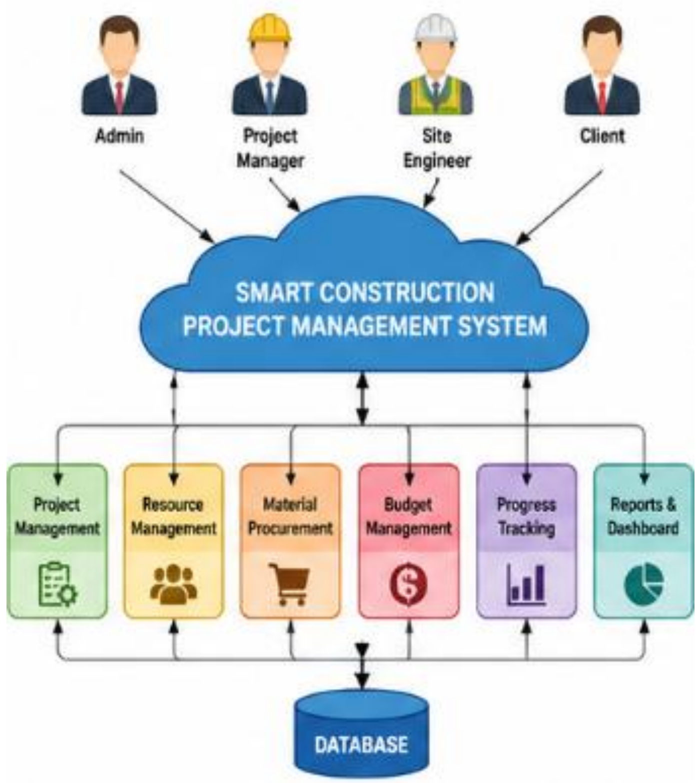

Figure 1. Smart Construction Project Management System Architecture

Illustrates the overall architecture of the proposed Smart Construction Project Management System, showing the interaction between users, functional modules, and the centralized database.

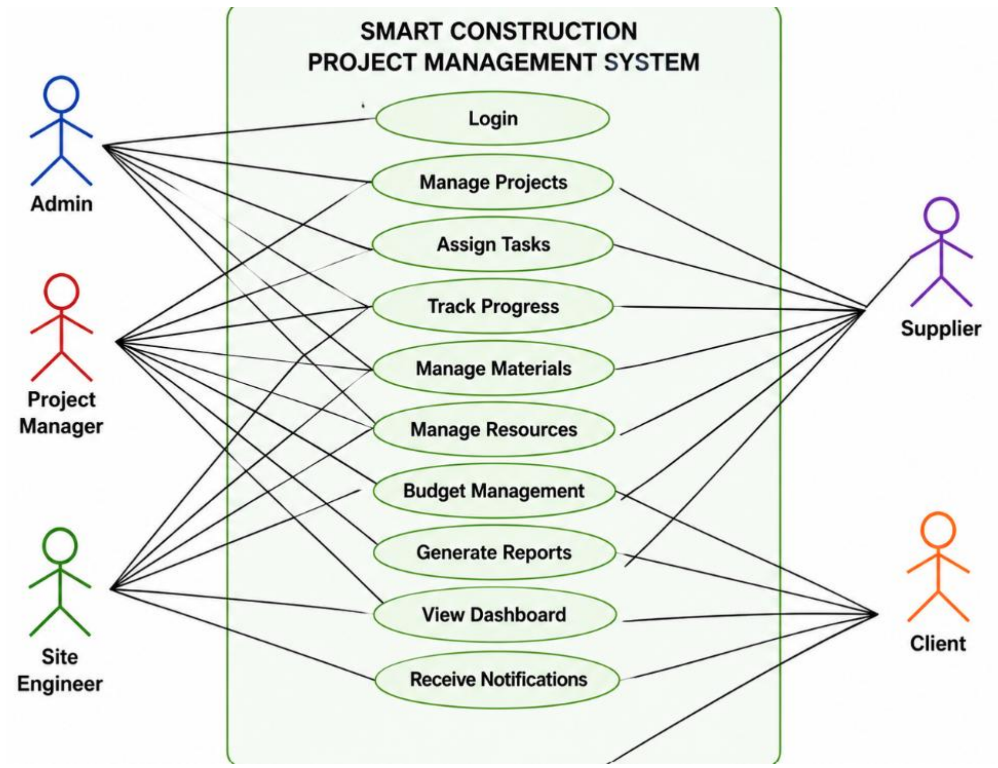

Figure 2: Use Case Diagram of the Smart Construction Project Management System

Shows the interactions between system stakeholders (Admin, Project Manager, Site Engineer, Supplier, and Client) and the major functionalities provided by the proposed system.

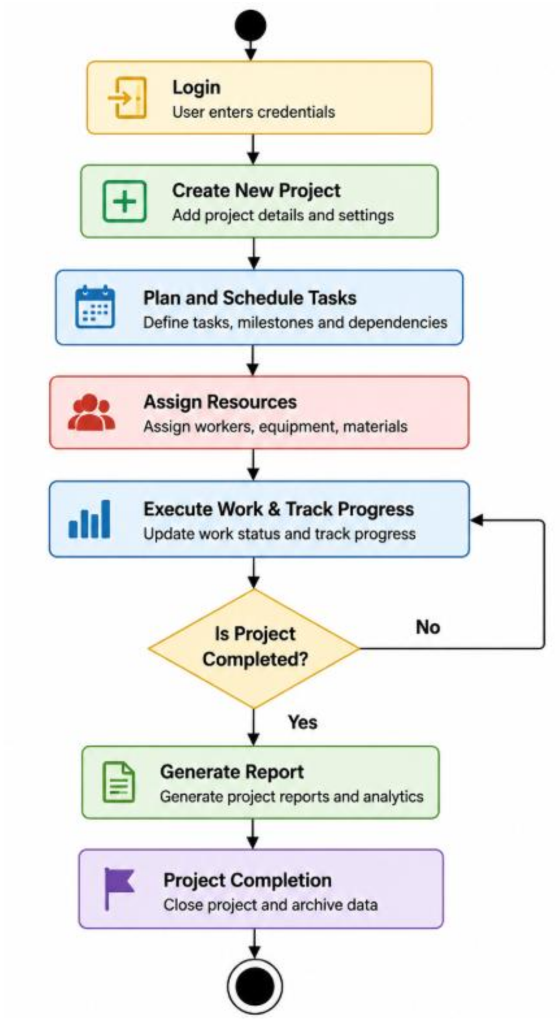

Figure 3: Activity Diagram for Smart Construction Project Workflow

Represents the workflow of construction project management, starting from project creation and task scheduling to progress monitoring, report generation, and project completion.

Figure 4: Functional Module Diagram of the Smart Construction Project Management System

Illustrates the major software modules of the proposed system, including project management, resource management, procurement, budgeting, reporting, notifications, and administration**.**

5. Justify the Chosen Methodology
Selected SDLC Model
Agile Scrum

Why Agile Scrum?

- Construction requirements may change during execution.
- Frequent stakeholder feedback improves the system.
- Faster software delivery through short sprints.
- Easy to manage changing customer requirements.
- Continuous testing improves software quality.

How Agile Supports Development

- Sprint Planning
- Daily Scrum Meetings
- Sprint Review
- Sprint Retrospective
- Continuous Integration
- Continuous Improvement

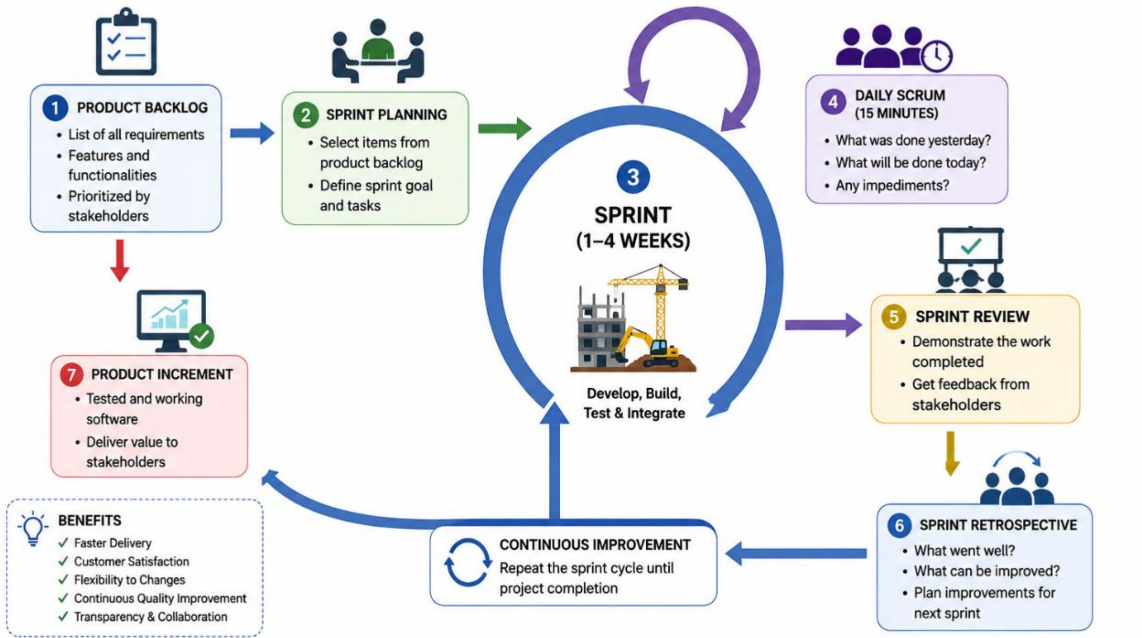

Figure 5: Agile Scrum Workflow for Smart Construction Project Management System Development

Depicts the Agile Scrum software development lifecycle adopted for the proposed system, highlighting iterative development through product backlog, sprint planning, implementation, review, and continuous improvement.

6.Expected Outcomes and Limitations

Expected Benefits

- Better project planning
- Faster communication
- Reduced delays
- Improved resource utilization
- Better budget control
- Real-time project monitoring
- Automatic report generation
- Higher customer satisfaction

How the Solution Addresses Challenges

- Centralized database eliminates duplicate records.
- Real-time tracking improves project visibility.
- Automated procurement reduces delays.
- Dashboards support better decision-making.
- Notifications improve communication.

Risks & Limitations

- Initial implementation cost
- Employee training required
- Internet dependency
- Data migration challenges
- Resistance to new technology

Future Enhancements

- AI-based delay prediction
- IoT sensors for construction monitoring
- Drone integration for site inspection
- Mobile application
- BIM (Building Information Modeling) integration
- Predictive analytics
- Cloud-based backup and disaster recovery

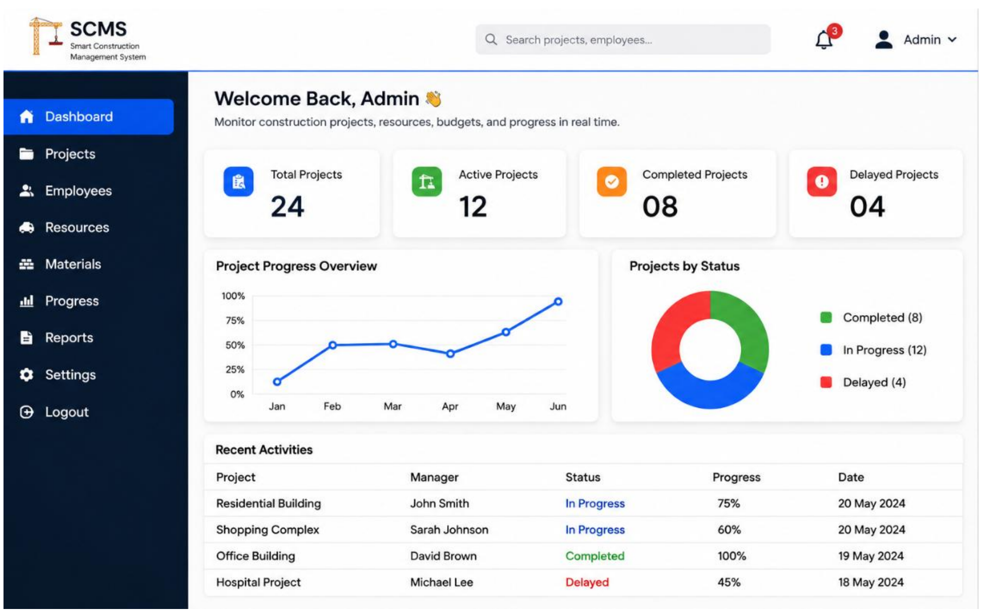

Figure 6: Dashboard Mockup of the Smart Construction Project Management System

Displays a sample dashboard providing real-time project information, including project status, budget utilization, task progress, reports, notifications, and key performance indicators for effective project monitoring.